Introduction to AGNTCY and the

Internet of Agents

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A paradigm shift in computing is underway

Deterministic → Probabilistic

[cloud computing → agentic computing]

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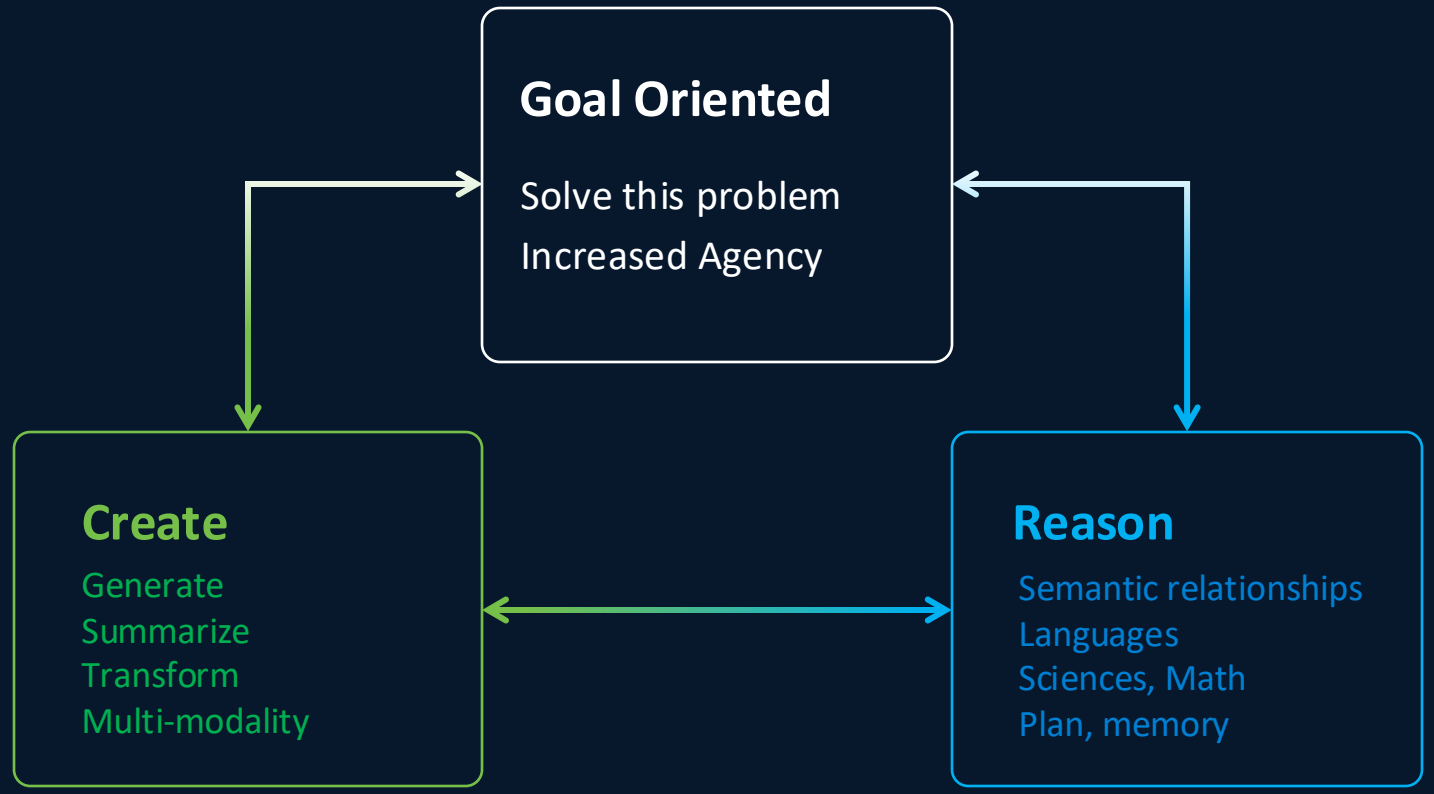
What’s different



Deterministic + Probabilistic Computing

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Agentic computing needs a collaboration framework

Agents

come from different **vendors**

use different development **frameworks**

running on different **clouds**

have different **access** controls

belong to different **organizations**

take on different **profiles** and **personas**

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Agents **must** collaborate to solve problems

Like a team of subject matter experts

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Focused on driving the Internet of Agents revolution

The Internet of Agents is an open[1], interoperable, internet for

quantum-safe

agent-agent collaboration

[1] An open, interoperable Internet of Agents will drive maximum value for all players: infra builders, operators, app developers, consumers

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Why do we need a new internet

Agents have **human-like attributes** and communication needs but at machine **speeds** and **scale**

1 discover and identify

- find agents for specific tasks? are they reputable?
- identify them, give access on my behalf?

2 securely connect

- assemble them to collaborate on the job to be done?
- interpret probabilistic / natural language inputs and outcomes?
- efficiently & securely transfer text-video-image-audio state?

3 observe and evaluate

- doing what they are supposed to do?
- getting into conflicts, loops

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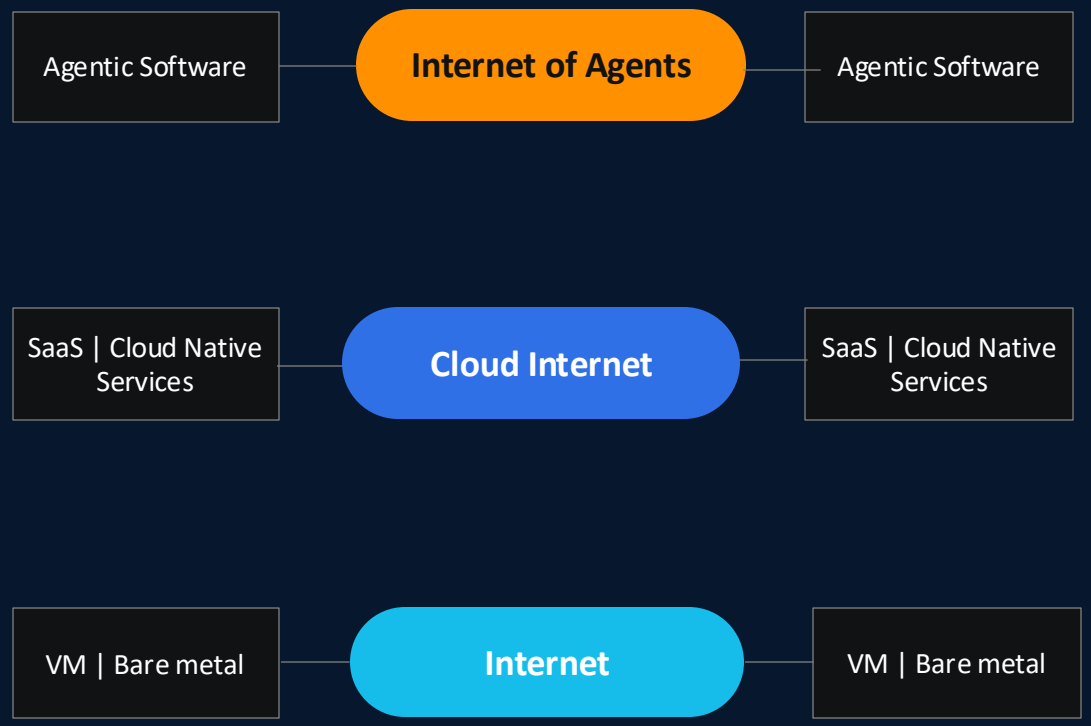
Rethinking the OSI model



semantic	β	Semantic Protocols
syntactic	α	Agent Connect Protocol Model Context Protocol
application	7	SLIM HTTPS, RPC, Mesh
presentation	6	SSL, TLS
session	5	BSD Sockets
transport	4	TCP, UDP, QUIC
network	3	IP, IPsec, OSPF
data link	2	Ethernet frames, ARP
physical	1	802.3 100G, DOCSIS

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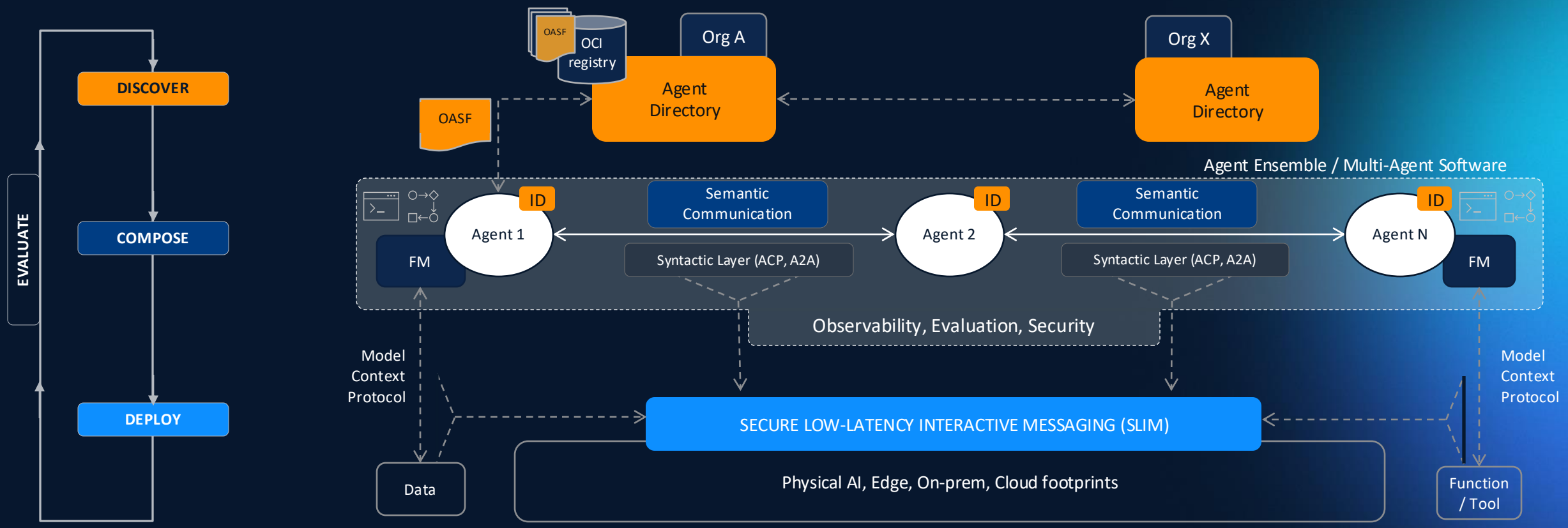
Internet of Agents architecture



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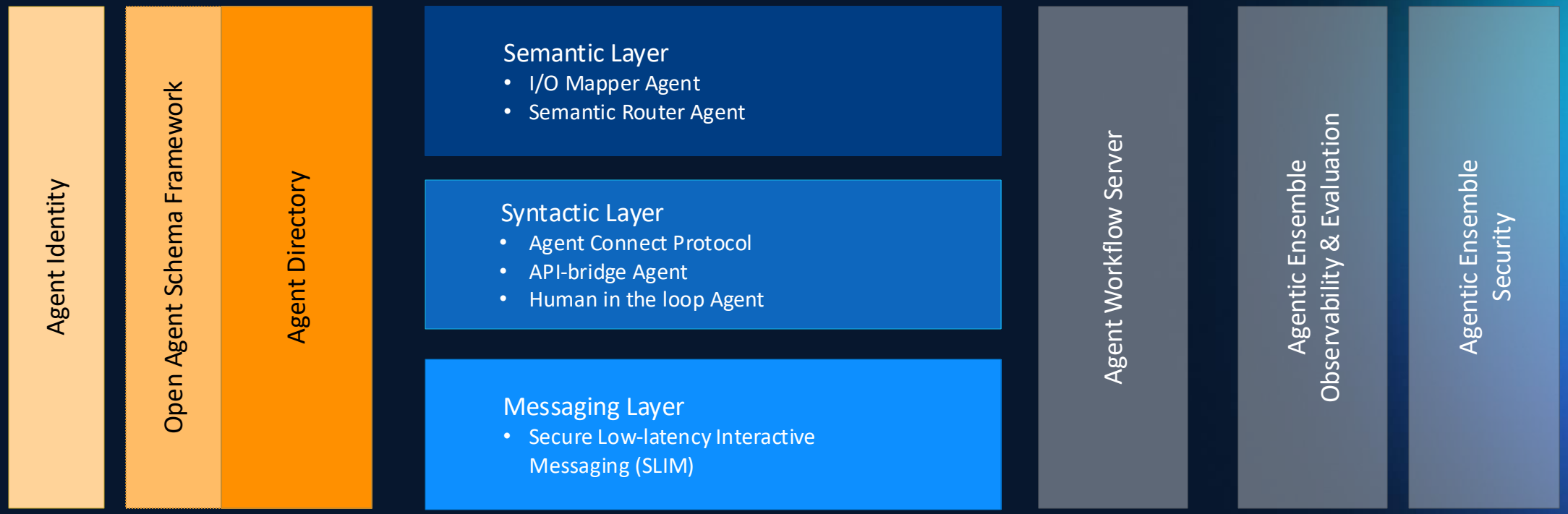
Internet of Agents stack



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Driving open standards and open source code

AGNTCY

An open source collective for inter-agent collaboration

The AGNTCY is where we are building the Internet of Agents to be: A diverse, collaborative space to innovate, develop, and maintain software components and services for agentic workflows and multi-agent software.

50+ collaborating organizations, growing…

STEERING COMMITTEE

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LangChain
Galileo

MEMBERS AND CONTRIBUTORS

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Agency
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ANON
anyweb
ArcBlock
aviz AI for Networks Networks for AI
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AGNTCY ❤ Protocols

MCP, A2A, … ACP

SLIM

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JARVIS

A multi-agent SRE with serious agency

20+ AGENT ENSEMBLE

- Support desk: 3 engineers ➡ JARVIS
- Query responses: Hours ➡ seconds
- 30% tasks: One day ➡ minutes

JARVIS ON CNOE GIT HUB

50+ TOOL & DATA CALLING

RAG: Platform docs

Live Tool Calling: K8sGPT, PagerDuty +

Graph RAG: AWS, Backstage, STO +

5+ USER INTERFACES

Webex chat, Backstage, VS Code, Jira, CLI +

10+ WORKFLOWS

Interact with JIRA

Many self-service functions

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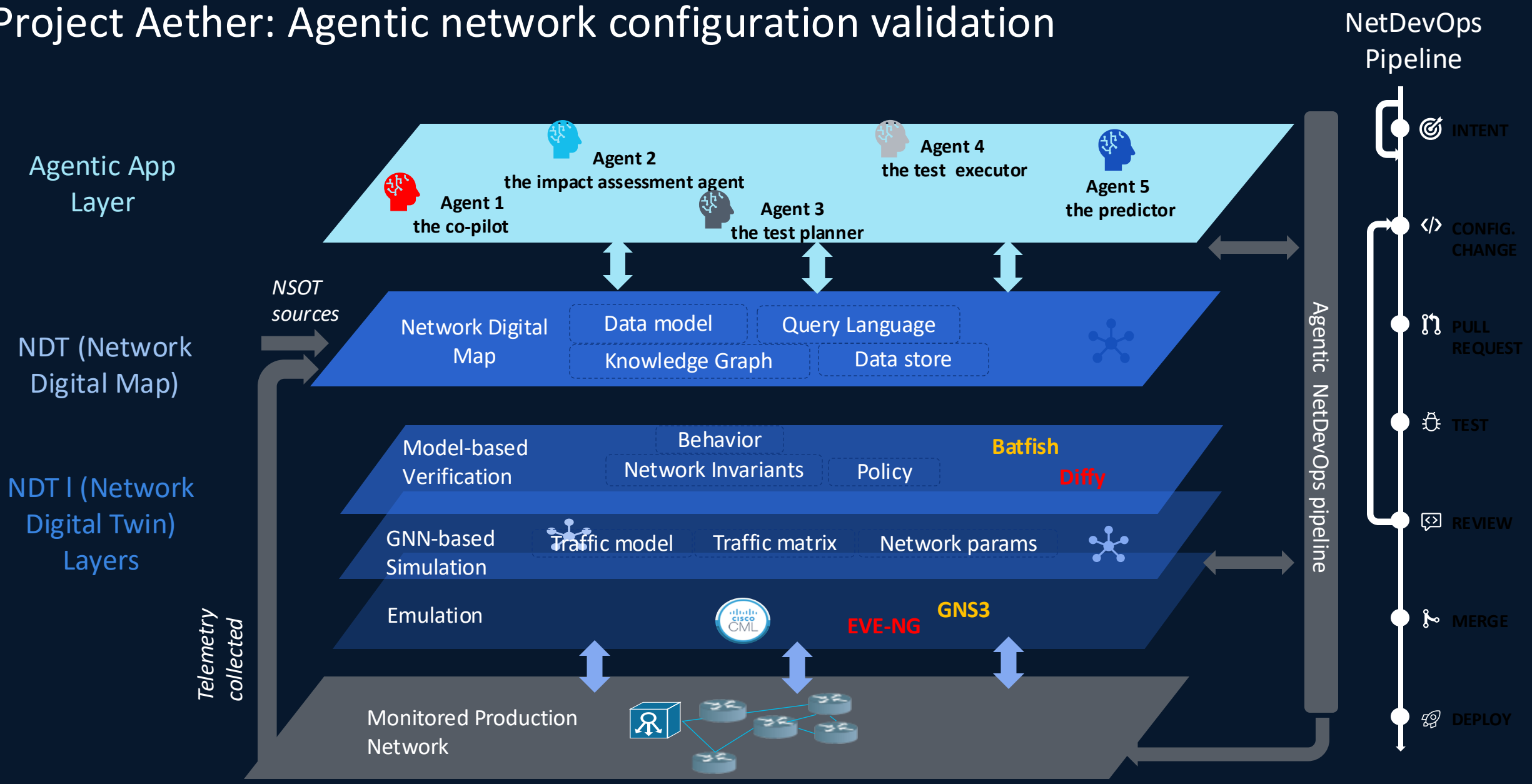
Project Aether: Agentic network configuration validation



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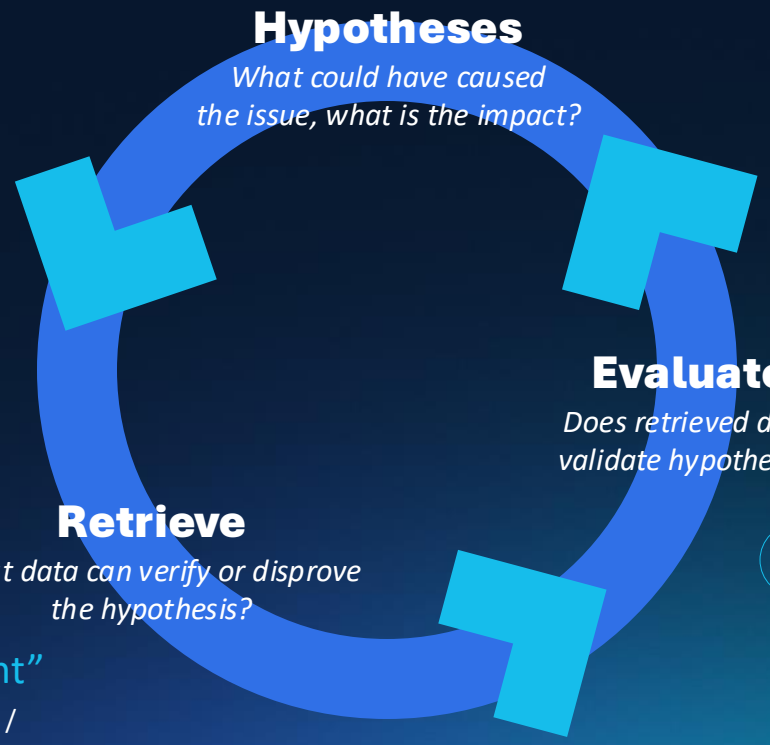
Project A3PO: AI Agents Troubleshooting like Humans



"Diagnosis Agent"
Classify the issue, create hypotheses on causes

"Retrieval Agent"
Capture relevant info / collect more info to support hypotheses evaluation

"Evaluation Agent"
Verify/disprove hypotheses, analyze dependencies between causes

"Analysis Agent"
Perform root-cause and impact analysis integrating the overall investigation

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Follow-up with these Cisco Live 2025 Sessions

The Internet of Agents: Building an Open and Interoperable Agentic AI Ecosystem - BRKETI-1009
Guillaume de Saint Marc

Agentic AI and NetDevOps: Enabling a New Era of Network Change Validation - BRKNWT-2502
Giovanna Carofiglio

A New Era of Cross-domain Diagnostics with AI and Agentic Systems - BRKETI-2172
Frank Brockners

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